

A working roadworks planning system

Built and running. Real Melbourne data, traffic simulation and AI road-user insights.

REAL DATA FOUNDATION OSM roads | SCATS traffic counts | PTV timetables | City pedestrian counts

GITHUB ↗

OPEN LIVE DEMO ↗

PROBLEM STATEMENT

Closures disrupt **drivers, public transport and pedestrians**. Planners need to test the impact, including **overlapping works**, before deployment.

OUR INNOVATION

Four AI driver personas

Commuter / Local / Visitor / Delivery

Four LLM driver personas read the same sign differently. Their readings inform the simulation.

AI interprets. The engine calculates.

VALUE & PATH TO MARKET

Help **councils and contractors** compare traffic impact and equipment cost before crews arrive.

Proposed model: RPM Hire pairs impact reports with hire quotes; councils use the evidence for permit review.

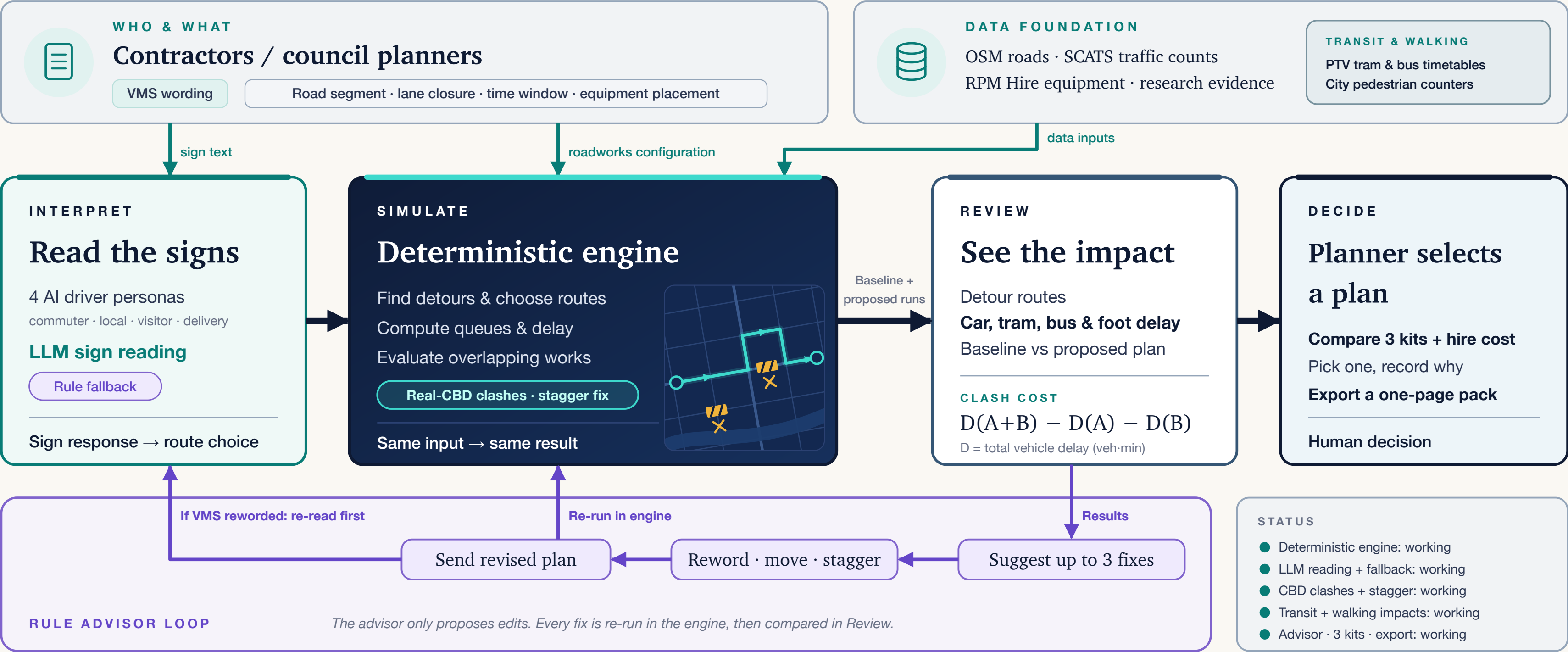
Next: pilot on one worksite.



Product view: La Trobe St, 5 pm: one closed lane is absorbed (0 m queue). At Lonsdale St, 8 am, the same closure queues **538 m even with a detour sign** (p.3). Map movements are illustrative; traffic impacts are computed by the engine.

System architecture

Under the hood: AI interprets. The engine calculates.



TECH STACK Python / OSMnx data pipeline, JavaScript simulation and Canvas UI, DeepSeek sign reading, Cloudflare Workers.

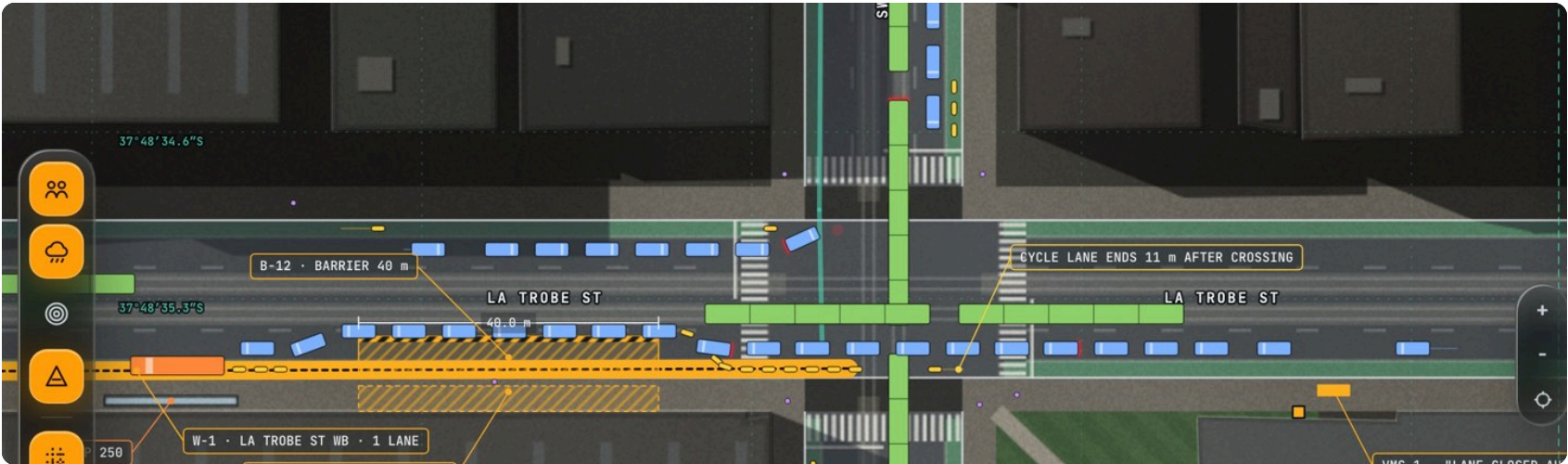
DAY 3 DELIVERABLES

1. Public code with a judge README.
2. One live CBD closure: set-up to exported execution pack.
3. An assumptions-and-sources sheet for every shown result.

A working system, end to end

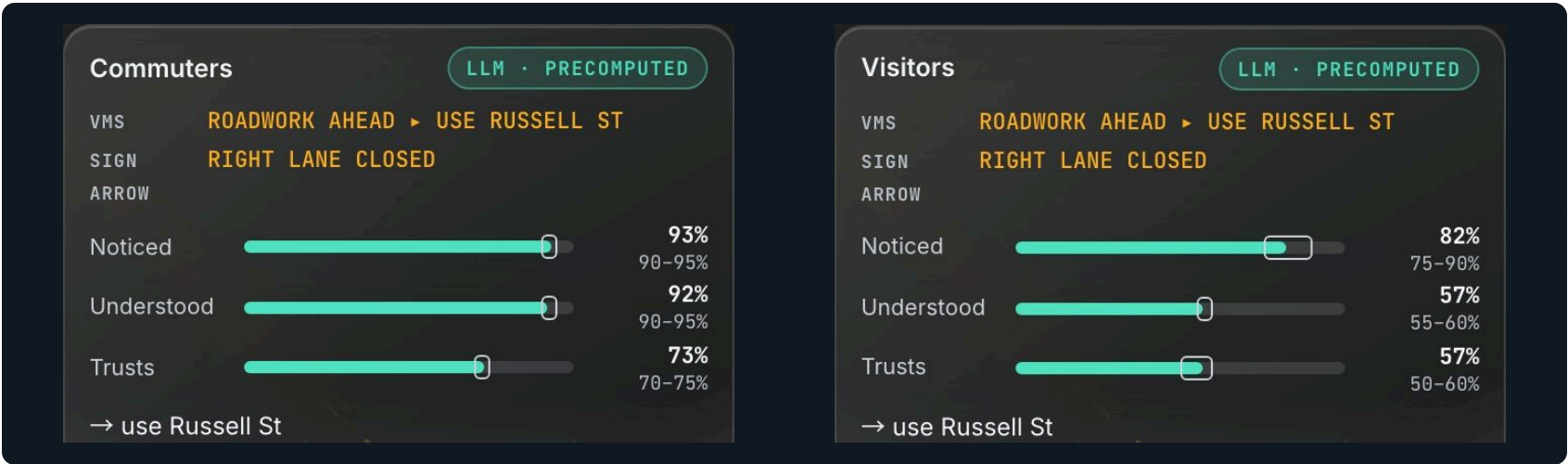
Real product screens. From a road closure to a plan you can act on.

02 Junction Sim · Cars and pedestrians



● Cars ● Pedestrians ● Cyclists ● Trams · Illustrative micro-simulation.

03 Impact · AI road-user perspectives



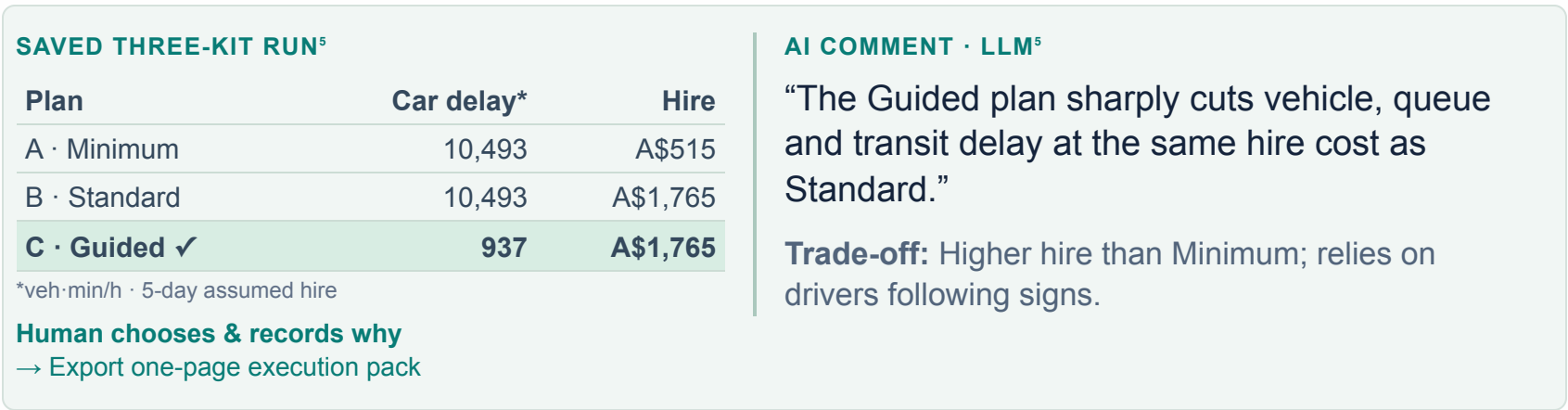
Commuter: clear directions. Visitor: unfamiliar streets, less confidence.⁴

03 Impact · Roads, transit and walking



Network ripple, public transport and access. Footpath-closure variant.³

04 Improve · AI comments → choose → export



Warning-only VMS / arrow boards give no route advice, so no extra diversion is predicted; **safety value is not scored.** C names the detour on the same A\$1,765 kit.

-46% vehicle delay: add “USE RUSSELL ST”¹

3-day shift ends the overlap between works²

+176 m walk if the south footpath also closes³

MODELLED
SCENARIO RESULTS

¹ Lonsdale WB, weekday 08:00: one lane closed, footpath open. Add “USE RUSSELL ST” to “ROADWORK AHEAD”: 10,493 → 5,629 veh·min/h (-46%); 538 m queue. ² Demo register: shift other works by 3 days to end overlap. ³ Same site, south footpath also closed: +176 m; ~185 ped/h, interpolated. ⁴ Precomputed sign readings; model behaviour. ⁵ Saved three-kit run, same site/hour, footpath open; LLM comment on those saved outputs. Assumed: passenger loads, driver mix/behaviour, stock and hire rates. Different VMS wording; results are modelled, not field measurements.